

# What are the most influential theories of cognitive development?

T/Th 9:35-10:25am

Professor Casey Lew-Williams

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# Midterm exam is on Thurs, Oct. 9

- Exam will be during our usual class time: 9:35-10:25am
- Content from lectures, textbook, precept, & precept readings is fair game.
- Do not memorize names of researchers or specific kid ages.
- If you have alternate accommodations for the exam, you'll hear from us.

# Journal article assignment

- See Canvas!
- **Your research topic is due on Canvas today/tomorrow** (follow your natural curiosity! Choose a topic related to your personal experiences/history/background.)
- **Research question** (don't make it huge or unrealistic!)
- **Hypothesis** (a proposed explanation - based on available evidence - that calls for further investigation)
- Conducting an article search
- Introduction (cite at least 3 scientific sources!)
- Method (this part will probably be the most enjoyable)
- Discussion (very important!)
- In-text citations
- References
- **In precept, you will develop grading guidelines, and we'll compile everybody's wording to create guidelines for the whole class.**

# Strange writing

“[Hello Kitty] inhabits the ‘thingness’ of the ‘thing’ in the physical properties of cuteness she brings to meaning making.”

*(Pink Globalization: Hello Kitty's Trek Across the Pacific, 2013)*



# Strange writing

A favorite quote from a student paper:

“This theory illustrates deficits in everything except some other things.”

# Strange writing

together—in and through multi-agency, multi-medial, and multi-channel processes of global dissemination. The problematization of security, fear, and danger that similarly tracks this new articulation of life begins to be thoroughly revised to the extent that it begins to revolve around this new account of the real—pursuing the logic of such a life of becoming into the global catastrophe of the becoming-dangerous of life to life itself. Where once Foucault interrogated the dynamics of the dangerous individual that followed from certain anatomo as well as biopolitics of life (Foucault 2004c), for contemporary biopolitics being *tout court* is becoming-dangerous.

This does not mean that subjectivities, boundaries, and territorialities, which once furnished the markers of certainty that helped to demarcate and operationalize traditional security regimes, lose all significance. It means instead that subjectivities, boundaries, and territory are comprehensively reconfigured as they are for example through the war on terror. They become a locale for the endless watch on the becoming-dangerous of an emergent life of becoming haunted by the evil integral to the very reasoning faculty by which it seeks technically to govern the conduct of its conduct while appealing for that revolution in disposition without which the omnipresence of evil will remain terrorizingly omnipresent. Significant, but differently, these become technically determined by the preoccupation with

# Strange emails

yessssssssssssssssssssss!!!!!!

THANK YOU!!!

you are my favorite professor! i am thinking about transferring my major to psychology(no i am kidding):D  
you have giant eyeballs!

—

I know you said no emails, but I just have a fewwww questions...

So if I learn all those words, define them and understand them and everything- will I have a chance at passing? Or is that just 1/100000000th of the test? >\_> Ive also been studying through the slides that youve given us from last exam to this exam...which is like 1000000000 stacks of paper...um, are there any chapters or slides that are especially important, that i should focus on?? I know I should have memorized all this stuff day by day, all semester, taking readable notes, and understanding each concept one by one...but my notes are crap and I have the attention span of a gold fish and I'm just not smart in general!!Well,atleast this time I plan on showing up, unlike the LAST exam (I spazzed out last minute and didnt go to class- real smart, I know)

okay I'm going to stop rambling now...back to the point....

whenever you have time please let me know(about the first part of this email)!



# GPT believes that theories matter. Do you?

why do theories matter?



Theories matter for several key reasons:

1. **Framework for Understanding:** Theories provide structured ways to interpret complex phenomena, offering a lens to make sense of the world. For instance, scientific theories like evolution or relativity explain natural processes in a way that helps us predict and understand future events.
2. **Guidance for Research:** Theories shape how researchers ask questions and conduct investigations. They provide a foundation for hypotheses and guide the methods used to test and validate ideas.
3. **Explaining Patterns:** Theories help identify and explain patterns or relationships that might otherwise go unnoticed. For example, economic theories explain market behavior, while psychological theories explore human behavior.
4. **Problem-Solving:** Theories offer solutions or frameworks for addressing practical problems. In engineering, medical science, or social policy, theories are applied to devise interventions, create technologies, or formulate strategies.
5. **Predicting Future Events:** Theories allow for predictions, helping us anticipate outcomes in various fields, from science to social interactions. The better a theory, the more accurate the predictions.
6. **Challenging Assumptions:** Theories encourage critical thinking by questioning the assumptions of existing knowledge. This fosters innovation and drives fields forward by opening up new lines of inquiry.
7. **Social and Cultural Impact:** Theories influence societal beliefs, values, and policies. Political and economic theories, for example, shape governments and institutions, affecting daily lives.

# Dynamic systems theory

- Focuses on how change arises from interactions between systems across timescales (both short and long).
- Example: **velcro mittens**
- Example: **language learning** emerges from cognitive, linguistic, socio-emotional, and environmental influences.

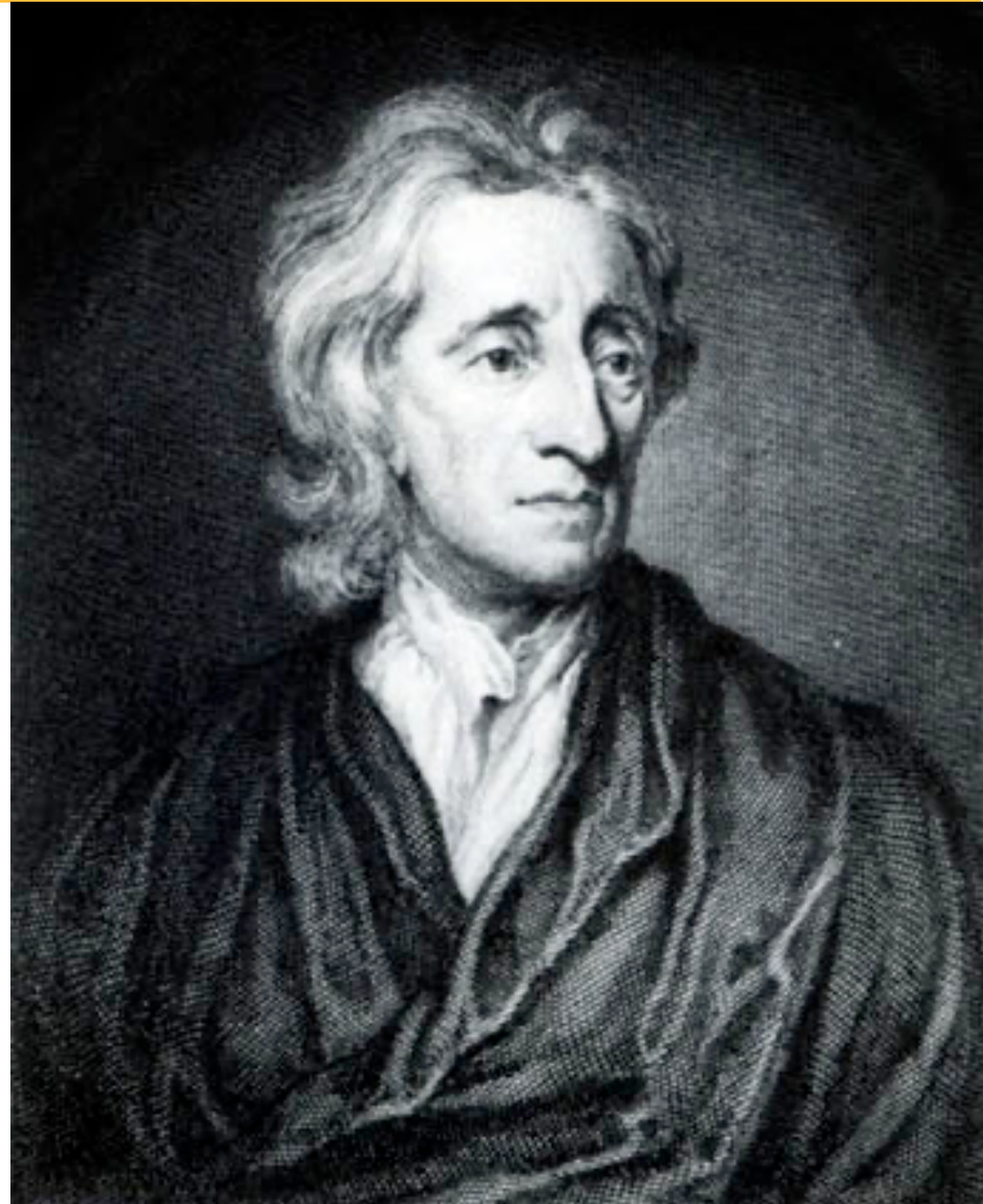
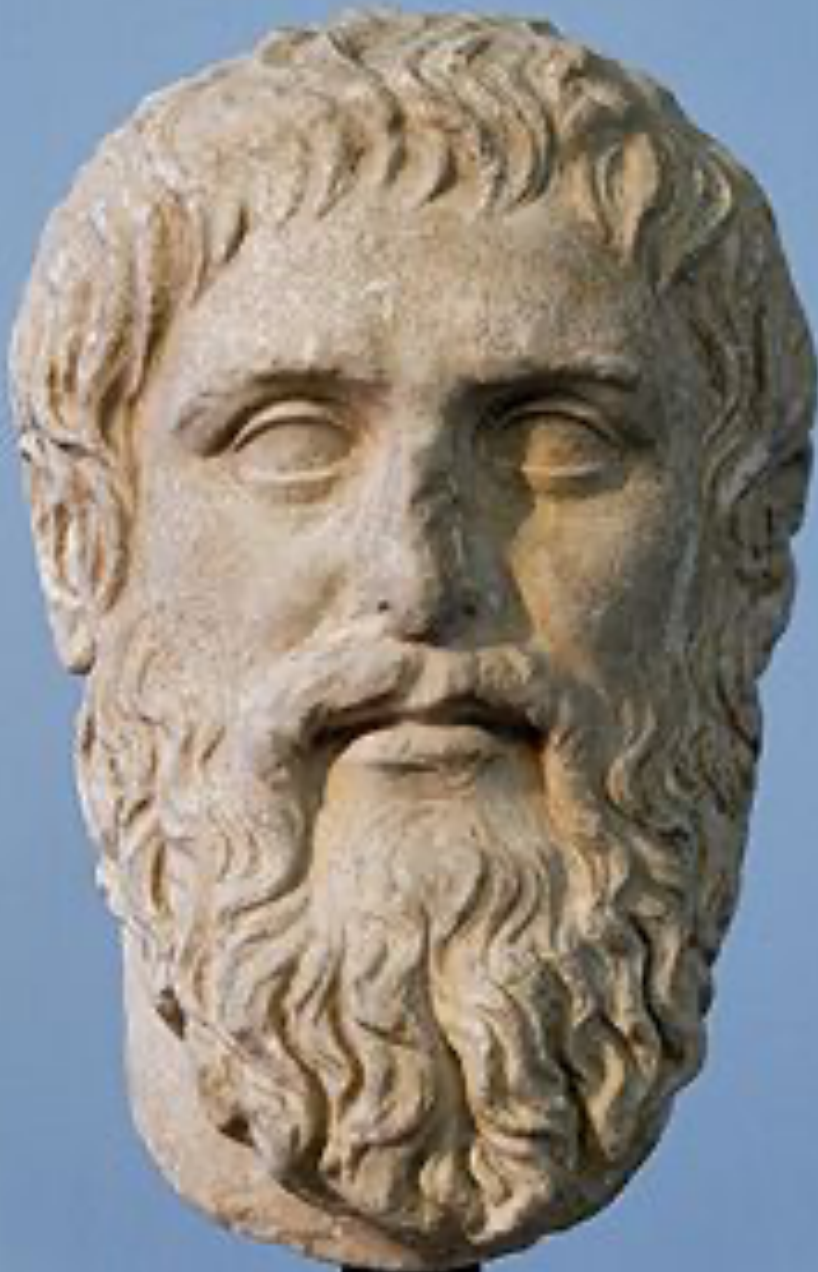




Jean Piaget  
1896-1980



**Plato** (5th century BCE) vs. **Locke** (1632-1704)



“Parents wonder why the streams are bitter, when they themselves have poisoned the fountain.” -Locke





# Jean Piaget



Children's intelligence tests

Focus on error

Became interested in the nature of children's thinking

# Two views of error

$$2 + 2 = 5$$

Random perturbation  
vs.

Systematic use of an alternative













start  
here

end  
here



# Constructivism

Children ‘construct’ knowledge on the basis of their experiences in the world. They are active learners from birth; their thinking and activities contribute greatly to their development.

“Laurent [Piaget’s infant child] is lying on his back... He grasps in succession a [number of objects], stretches out his arm, and lets them fall. He distinctly varies the position of the fall. When the object falls in a new position, for example, on his pillow, he lets it fall two or three more times on the same place, as though to study the spatial relations.”



# Change mechanisms according to Piaget



## 1. **Assimilation**

Children translate new information into a form they can understand. They fit something new into their current theories.



Pig!  
Bear!



# Change mechanisms according to Piaget



## 2. **Accommodation**

Children improve their current understanding in response to new experiences. Sometimes this means they create new theories.



Wombat!

Smaller than bears.

Furrier than pigs.

Live underground.

Marsupials.

# Change mechanisms according to Piaget



## 3. **Equilibration**

Using both assimilation and accommodation to create a stable understanding of something.

Equilibrium: satisfied with understanding.

Disequilibrium: perceive something is wrong.

Equilibrium: more sophisticated understanding.

Example in a 6-year-old:

1. "Animals are the only living things."
2. But plants grow and die. Uh-oh. [disequilibrium]
3. New concept of life (child moves from *disequilibrium* to to *equilibrium* via *accommodation*)

# Piaget's stage theory



**Sensorimotor:** birth to 2 years

**Preoperational:** age 2 to 6/7 years

**Concrete operational:** age 6/7 to 11/12 years

**Formal operational:** to adulthood



# Sensorimotor: birth to 2 years



- Infants live primarily in the here-and-now
- Infants have basic motor skills, sensory/perceptual systems
- They adapt to and explore environment using assimilation, accommodation, and equilibration

# **FAIL:** object permanence



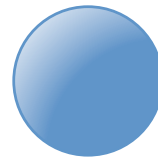
Object Permanence



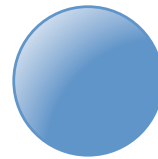
“Encourages developing motor skills and provides early lessons in cause and effect and object permanence.”



**FAIL:** A-not-B



**FAIL:** A-not-B



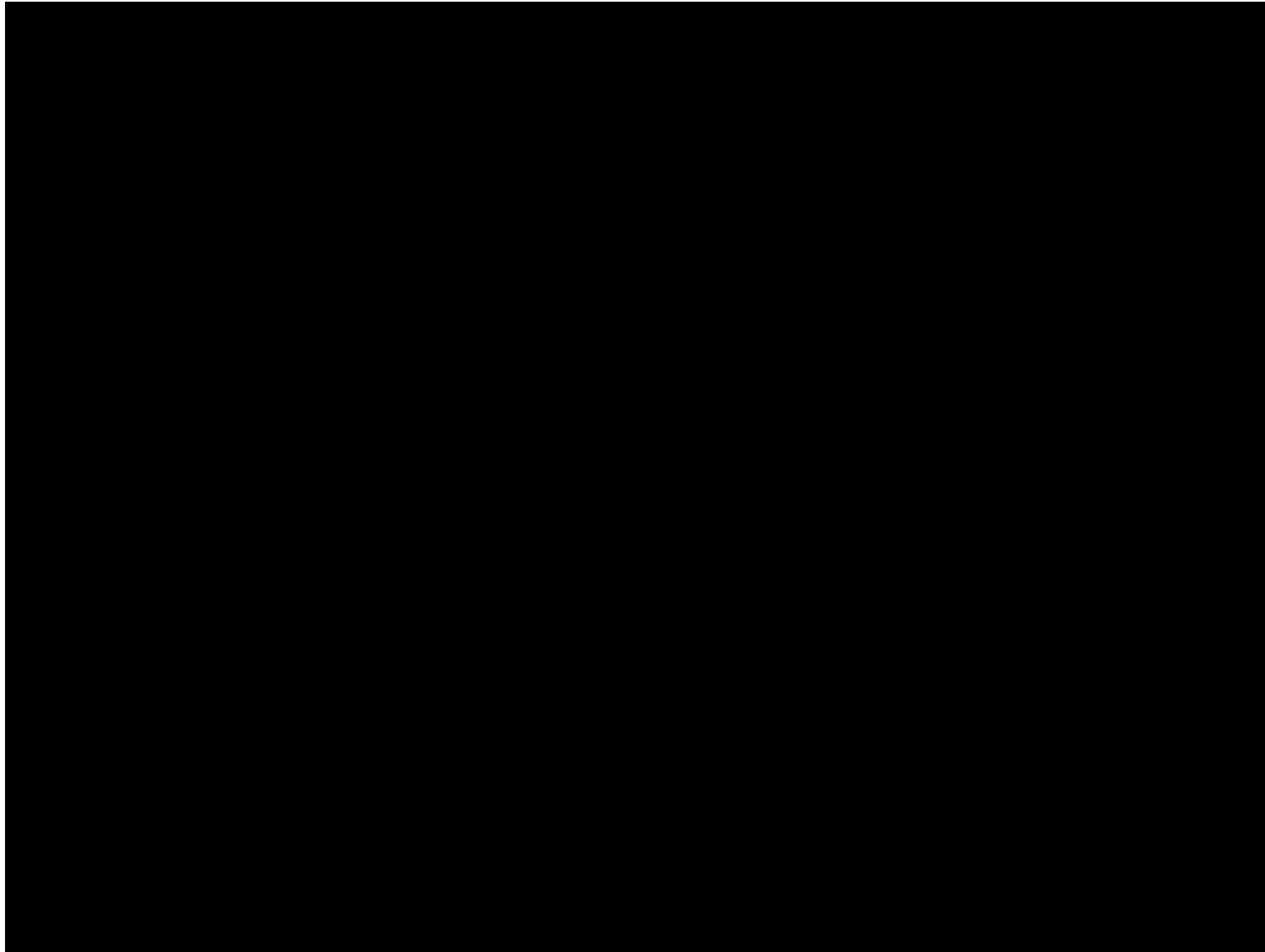
# Preoperational: age 2 to 6/7 years



- Toddlers begin to represent experiences in language, imagery, and symbolic thought
- Cannot perform “operations” (reversible mental activities)
- Focus on a single, perceptually salient aspect of an event (called centration)



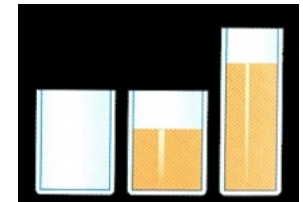
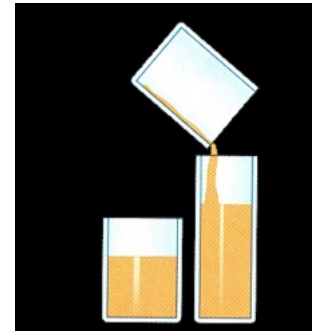
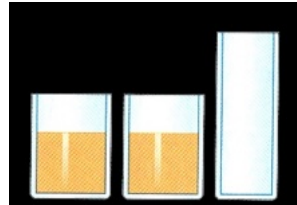
# **FAIL:** Conservation



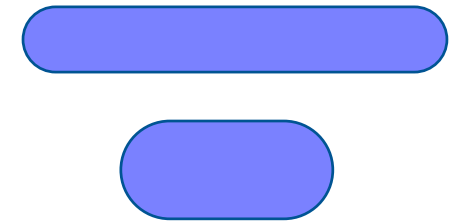
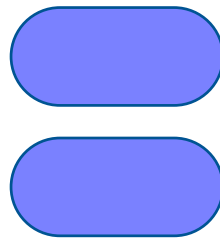
# FAIL: Conservation



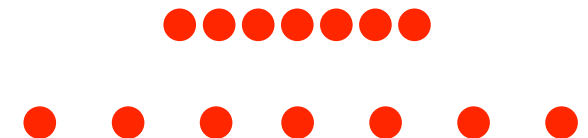
Liquid quantity



Solid quantity



Number



# FAIL: Egocentrism



Piaget's Three-Mountains Problem



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# FAIL: Appearance/reality



What is it?

“Rock”

Now touch it. What is it?

“Sponge”

What does it look like?

“Sponge”



# Concrete operational: age 6/7 to 11/12 years



- Children can reason logically about concrete objects and events
- But they are limited to concrete situations and have difficulty thinking in abstract terms when combining information systematically

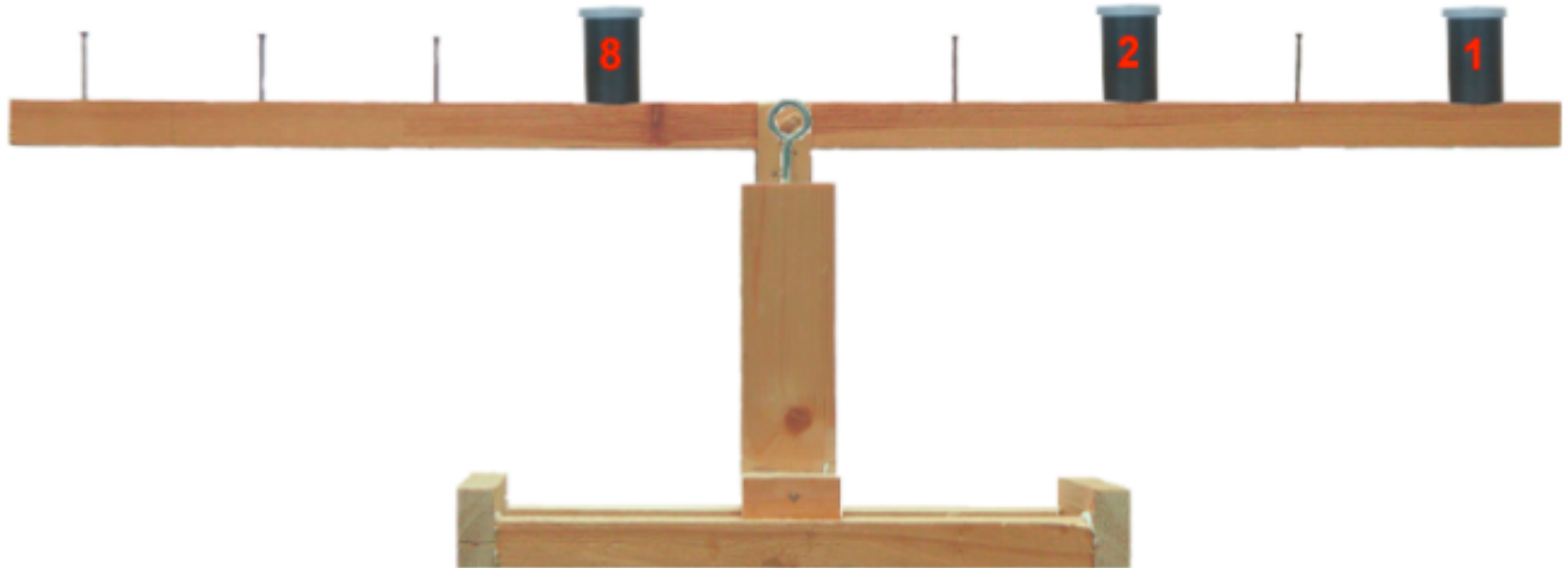
# FAIL: Formal logic



- All the bears in the North Pole are white.
- Eva saw a bear in the North Pole.
- What color was the bear that Eva saw?
- Adult: “White.”
- 9-year-old: “I dunno I wasn’t there.”



# **FAIL:** Systematic testing



# Formal operational: 11/12 years +



- Can think about abstractions and hypotheticals
- Can perform systematic “experiments” to draw conclusions about the world



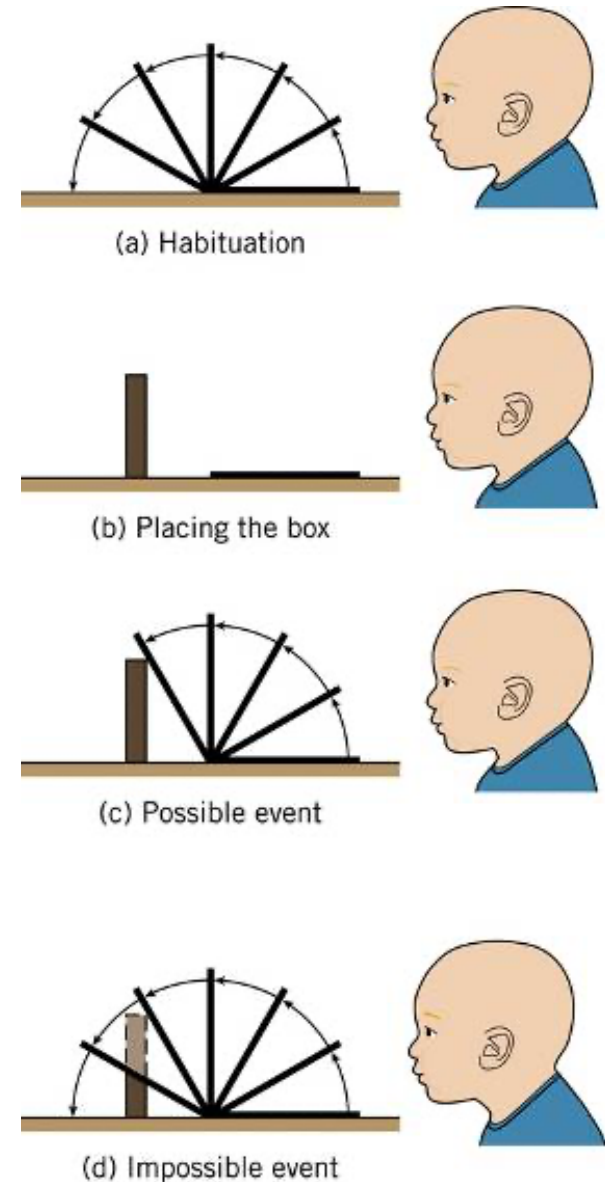


# Why is Piaget's theory impressive?

- It tries to explain so much!
- Most theories in the sciences are narrower
- Major impacts on education:
  - His work fostered learning environments that let children explore and discover on their own timeline
  - Child-centered
  - Hands-on learning and problem-solving

# Challenges to Piaget's theory

- Underestimated young children's abilities?
- Violation of expectation method: babies look longer at 'impossible' events.



**Many more  
theories**

# Core knowledge theory

- Humans are innately endowed with a cluster of ‘core systems’: knowledge related to important aspects of our evolutionary history.
- These systems represent abstract properties such as numerosity, objecthood, agency / goal-directed action, spatial navigation, & other people’s thinking.
- Akin to perceptual systems in being ‘modular’
- We do have domain-general learning abilities, but we also have innate mental structures that support quick acquisition of core knowledge.



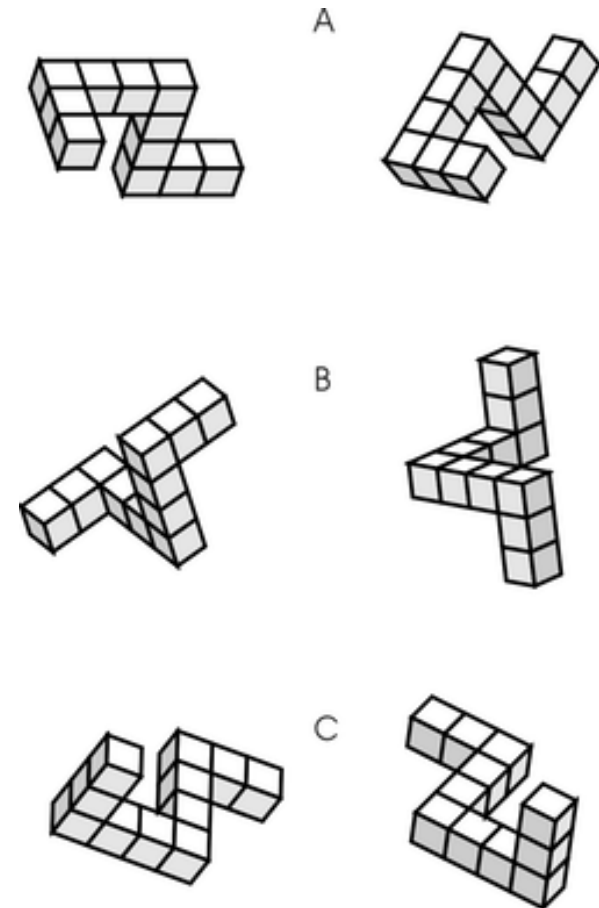
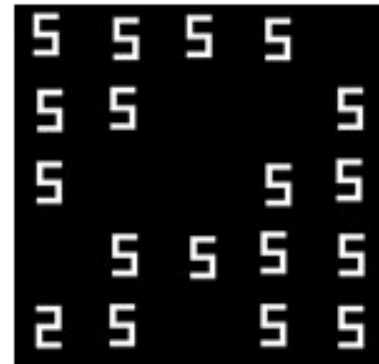
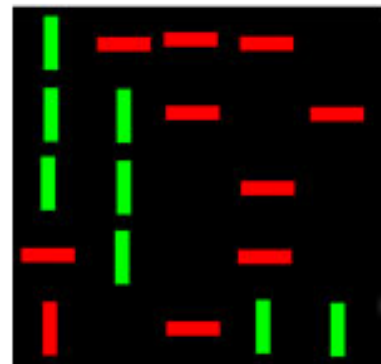
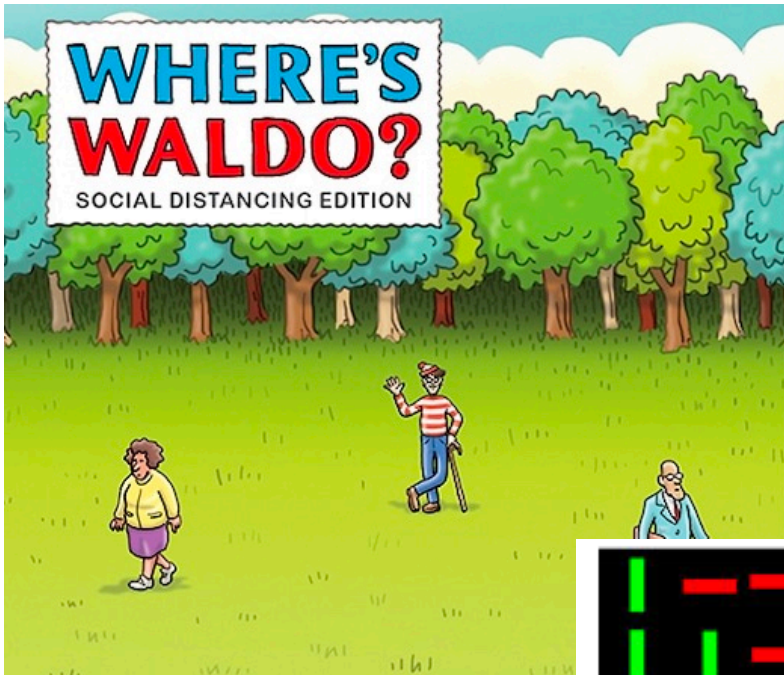
# Core knowledge theory

“Hearts evolved to support the process of blood circulation, livers evolved to carry out the process of toxin extraction, and mental structures evolved to enable the learning of certain types of information necessary for adaptive behavior”

(Gelman & Williams, 1998, p. 600)

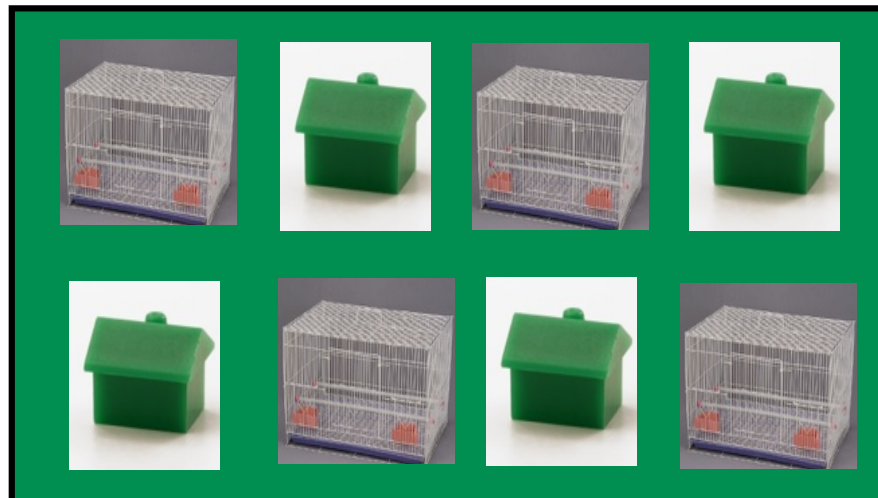
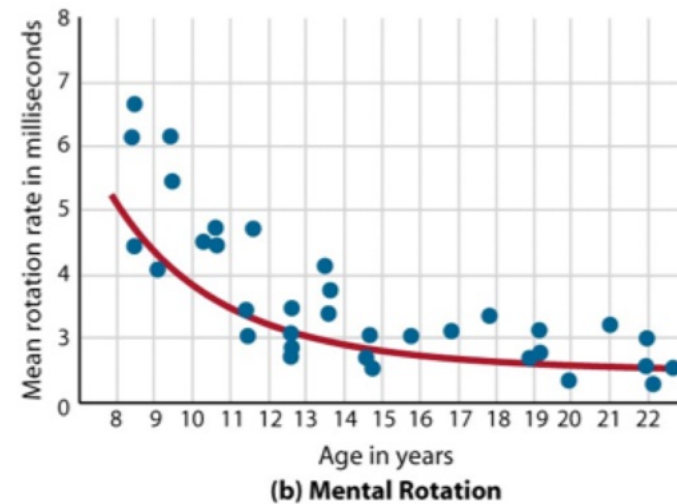
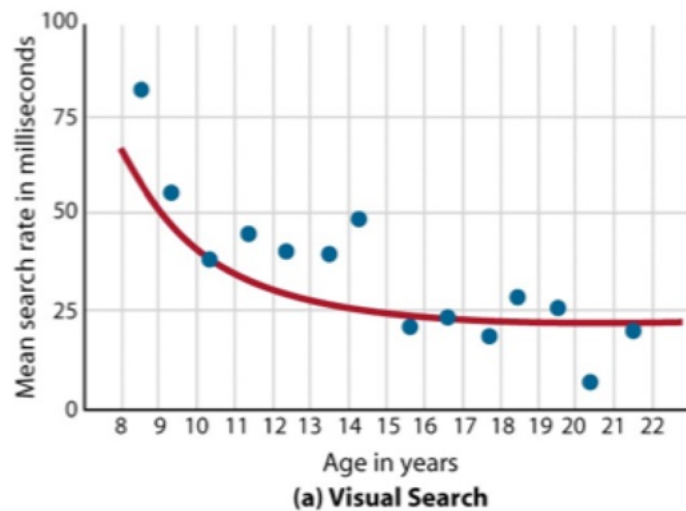
# Information processing theory

- Emphasizes input, processing, storage, and output. The mind is fundamentally a computational system.
- Focus on basic components of cognition: **attention**, **processing capacity**, **learning**, **memory**.
- Examples: visual search, mental rotation



# What develops in information processing theory?

- Processing speed
- Better strategies
- Executive function



# Sociocultural theory



- Lev Vygotsky (1896-1934)
- Cognitive development is a socially mediated process.
- Development occurs in social interaction; can't separate social environment from the individual.
- Culture affects learning.

# Sociocultural theory

- **Intersubjectivity:** development is not an independent activity, there is a shared understanding among participants in an activity.
- **Zone of proximal development:** distance between what a child can do independently vs. with help
- People learn to function in *their* communities, to think in ways that suit *their* goals, to solve problems present in *their* own environments





# Basketball + social touch

- Coded touch between NBA players during 2008-2009 season
- Controlled for player status, preseason expectations, & early season performance
- **Early season touch predicted improved performance for individuals *and* teams later in the season.**



# Ecological systems theory (or 'biocological model')



- Urie Bronfenbrenner (1917-2005)
- The child exists within nested layers of influence: micro, meso, exo, macrosystems
- Development happens through interactions between these layers.

